



NCCER Module 22102



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Heavy Equipment Safety





Section Two

Hazard Communication

Objective

2. Explain hazard communication.
 - a. Explain the purpose and content of safety data sheets (SDS).
 - b. Describe the labeling systems used for hazardous materials.
 - c. Describe noise control methods.



Performance Task

Under the supervision of an instructor, trainees should be able to do the following:

There are no Performance Tasks associate with this section.



2.0.0 Hazard Communication

- This section outlines the hazards beyond roll-over injuries and collisions.
- You may also be exposed to many types of chemicals used in products, such as diesel fuel, hydraulic fluid, and cleaning solvents.
- There is also risk from exposure to diesel exhaust and noise from the equipment.



2.1.0 Safety Data Sheets

- OSHA and MSHA require that all hazardous chemicals at a workplace be clearly identified.
- A file containing the SDSs for any potentially harmful materials should be located near the work area or in a supervisor's office.
- All SDSs should contain 16 sections as specified by OSHA.



2.1.1 Section 1—Product and Company Identification

- Proper name(s) of the product
- Chemical family that the product is associated with
- Name of the manufacturing company
- Phone numbers for contacting the manufacturer



2.1.2 Section 2—Hazards Identification

- Emergency overview of the product
- Appearance
- Combustibility
- Actions to avoid, such as inadvertent suction of the liquid into the lungs.
- Physical reactions you may experience if the product is inhaled or ingested, or if it comes in contact with skin or eyes.
- If the product is a carcinogen (causes cancer)

Potential allergic reactions may not be listed.



2.1.3 Section 3—Composition

This section identifies the hazardous ingredients that make up the product and lists the maximum exposure allowed in order to prevent skin damage.



2.1.4 Section 4—First Aid

- First-aid actions to take if the product is inhaled, ingested, or splattered onto skin or into eyes.
- These actions are the minimum needed. Company policies may dictate additional actions or requirements.
- *OSHA CFR 1926.50* requires that employers provide first-aid kits at every job site.



2.1.5 Section 5—Fire-Fighting Measures

- Specific hazards associated with the product
- Products recommended for extinguishing any fire involving the product
- Special PPE for fighting a fire fueled by the product

WARNING

In some situations, there are certain substances that should not be used on fires.

Make sure to review such warnings before attempting to extinguish any fire.



2.1.6 Section 6—Accidental Release Measures

- The Accidental Release Section of the MSDS addresses what to do if a product is accidentally released by way of spills, leaks, or unintentional releases.
- This section may list specific products recommended for absorbing the spilled or leaked product.



2.1.7 Section 7—*Handling and Storage*



- Proper storage decreases the risk of explosion and fire.
- Safety cans are used for safe transportation and storage of flammable materials. They are often color coded for different materials.
- Safety cabinets are designed specifically to reduce the risk of explosion and fire.

2.1.8 Section 8—Exposure Controls/ Personal Protection

This section lists specific PPE and how to use it.



(A) SELF-CONTAINED BREATHING APPARATUS (SCBA)



(B) SUPPLIED AIR MASK



(C) FULL FACEPIECE MASK



(D) HALF MASK



2.1.9 Section 9—Physical and Chemical Properties

- Appearance
- Solid, liquid, or gas
- Boiling point
- Melting point
- Flash point
- How the product reacts to different temperatures and pressures



2.1.10 Section 10—Stability and Reactivity

- Information about how stable the product is at given temperatures and pressures.
- Materials to keep separated from the product.
- Conditions to avoid that could cause the product to ignite or explode.



2.1.11 Section 11—Toxicological Information

- Toxicity of the product
- Details about the chemical reactions the product has had on both humans and test animals.

Be sure to read and understand the summary statements associated with the product.



2.1.12 Section 12—Ecological Information

Section 12 lists the environmentally damaging effects of the product. Diesel fuel, for example, can be **toxic** to the environment, animal habitats, and marine life.



2.1.13 Section 13—Disposal Considerations

- Explains how to clean up and dispose of a given product.
- Check with federal, state, and local guidelines to determine the proper disposal methods.



2.1.14 Section 14—Transport Information

- Different countries, and different states within the United States, have different guidelines on how to ship or transport given products.
- Section 14 of an SDS shows the product's proper shipping name, its hazard class, and how it should be packed for transport.
- The shipping vehicle must be labeled with the applicable UN/Identification number so that anyone can see the hazard level of the product being transported.



2.1.15 Section 15—Regulatory Information

- Regulatory guidelines that may be associated with the product
- Right-to-Know regulations of different states and communities



2.1.16 Section 16—Other Information

Section 16 is a catch all section. It may contain additional information that the product supplier wants to add.



2.2.0 Material Labeling

- National Fire Protection Association (NFPA) system
- Hazard Material Identification System

NAME OF MATERIAL	
3	HEALTH
4	FLAMMATORY
2	PHYSICAL HAZARD
B	PROTECTIVE EQUIPMENT

4	SEVERE HAZARD
3	SERIOUS HAZARD
2	MODERATE HAZARD
1	SLIGHT HAZARD
0	MINIMAL HAZARD
*	CHRONIC HAZARD
1	SLIGHT HAZARD
0	MINIMAL HAZARD



HEALTH	
4	DEADLY
3	EXTREME DANGER
2	DANGEROUS
1	SLIGHT HAZARD
0	NO HAZARD
REACTIVITY	
4	MAY DETONATE
3	EXPLOSIVE
2	UNSTABLE
1	NORMALLY STABLE
0	STABLE
SPECIFIC HAZARD	
OXY	MAY DETONATE
ACID	EXPLOSIVE
ALK	UNSTABLE
COR	NORMALLY STABLE
W	STABLE
☢	RADIATION

2.3.0 Noise

- When hearing protection devices are used, workers may not be able to hear safety signals.
- Long-term exposure to noise can cause noise-induced hearing loss or a permanent ringing in the ears, a condition called tinnitus.
- Earmuffs, ear canal caps, and earplugs are the three principal types of hearing protection devices (HPD).
- To protect hearing, avoid sources of loud noise as much as possible. Stay away from the area if not working there.



Wrap Up – Trade Terms

Hazard Communication (HazCom): An OSHA standard that requires contractors to educate employees about working with or near hazardous chemicals.

Toxic: The degree to which a substance can harm humans or animals.



2.0.0 Section Review Question 1

1. The SDS for diesel fuel recommends the use of an air-purifying respirator with filter that is black or _____.
- a. green
 - b. purple
 - c. red
 - d. yellow



2.0.0 Section Review Question 1 Answer

1. The SDS for diesel fuel recommends the use of an air-purifying respirator with filter that is black or _____.
- a. green
 - b. purple
 - c. red
 - d. yellow**



2.0.0 Section Review Question 2

2. A reactivity rating of 3 on a material label indicates that the substance ____.
- a. is stable
 - b. is unstable
 - c. is explosive
 - d. may detonate



2.0.0 Section Review Question 2 Answer

2. A reactivity rating of 3 on a material label indicates that the substance _____.
a. is stable
b. is unstable
c. is explosive
d. may detonate



2.0.0 Section Review Question 3

3. The noise threshold that establishes the need for double hearing protection is _____.
- a. 50 dBA
 - b. 60 dBA
 - c. 85 dBA
 - d. 105 dBA



2.0.0 Section Review Question 3 Answer

3. The noise threshold that establishes the need for double hearing protection is _____.
- a. 50 dBA
 - b. 60 dBA
 - c. 85 dBA
 - d. **105 dBA**



Next...

Section 3.0.0
Heavy Equipment Safety



*Read Sections 3.0.0 through 3.3.8 and
complete the Section Review.*